

8 Mitosis and the cell cycle are involved in producing new cells.

(a) Describe the role of the cell cycle in producing new cells.

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(b) Uncontrolled cell division can result in the formation of a tumour.

The RB gene is a tumour suppressor gene.

The RB protein produced from this gene inhibits tumour growth.

Lung cancer can occur when this gene has increased methylation compared with the RB gene in normal lung cells.

Explain why increased methylation of this gene could cause the growth of a tumour.

(4)



(c) The cell cycle can be affected by various chemicals.

One drug, used to treat various types of cancer, prevents the shortening of spindle fibres.

Explain how preventing the shortening of spindle fibres affects mitosis.

(2)

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*(d) Paclitaxel is a drug used to treat some tumours.

In one study, the effect of paclitaxel on the cell cycle in human lung tumours was investigated.

The table shows the results of this study.

Concentration of paclitaxel / nmol dm^{-3}	Duration of mitosis / hours	Mitotic index of cancerous cells in lung tissue (%)
0	1	2
10	4	17
100	9	38
1 000	21	48
10 000	24	59

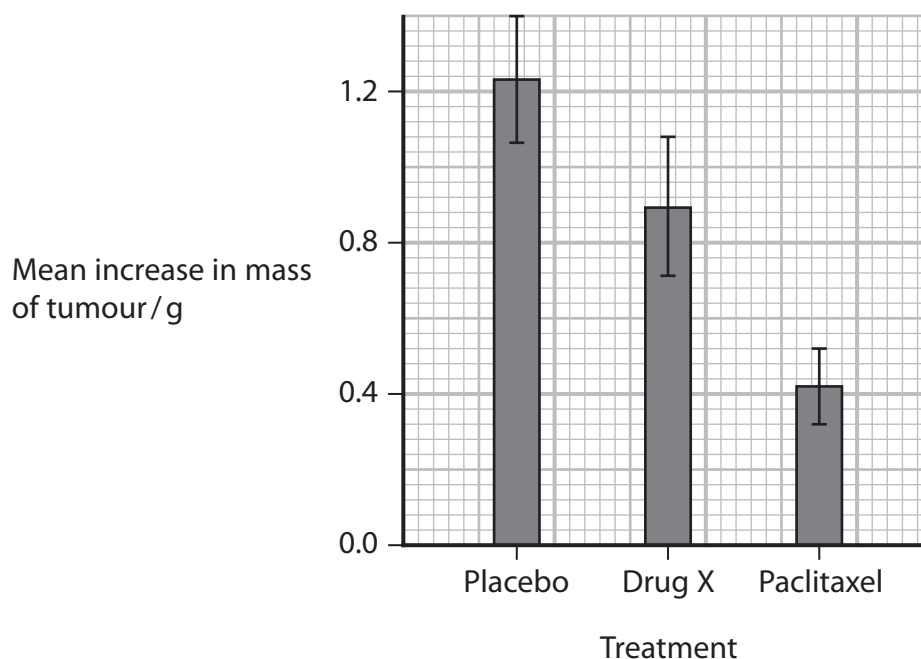
In another study, the effect of paclitaxel on the growth of breast tumours in mice was investigated. The effect of paclitaxel was compared with drug X and with a placebo.

For each group the increase in mass of the tumour was measured 21 days after starting treatment.

Three groups of mice with breast tumours were given the following treatments:

- Group 1 received a placebo
- Group 2 received drug X
- Group 3 received paclitaxel.

The graph shows the results from this study.



Evaluate the effectiveness of paclitaxel.

Use the information given to support your answer.

(6)

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(Total for Question 8 = 15 marks)

TOTAL FOR PAPER = 80 MARKS

